

COMPUTER NETWORKS LABORATORY

EXPERIMENT – 14

HOW TO RUN AN NMAP NETWORK SCAN

Nmap Network Scanning – Process

1. Aim

To perform a basic network scan using Nmap and identify accessible ports and services on an authorized target system.

2. Introduction

Nmap (Network Mapper) is a network scanning tool used to discover hosts, identify open ports, and obtain basic information about network services. It is commonly used for network administration, troubleshooting, and authorized security testing.

3. Safety Note

Only scan systems or networks that you own or have explicit permission to test. For laboratory practice, use your own computer, a local virtual machine, or an authorized lab target.

4. Requirements

- Nmap installed on the computer
- Command Prompt, PowerShell, or Terminal
- An authorized laboratory target IP address

5. Install and Verify Nmap

Ubuntu/Debian:

```
sudo apt update
sudo apt install nmap
```

Verify installation:

```
nmap --version
```

On Windows, install Nmap using the official installer and then open Command Prompt or PowerShell.

6. Find the Target IP Address

Windows: Open Command Prompt and run:

```
ipconfig
```

Linux: Open Terminal and run:

```
ip addr
```

7. Run the Basic Nmap Scan

Open Command Prompt, PowerShell, or Terminal and enter the following command. Replace **TARGET_IP** with the IP address of your authorized lab target.

```
nmap TARGET_IP
```

Example:

```
nmap 192.168.1.10
```

The scan reports information about accessible ports on the target.

8. Run Service / Version Detection

To identify the services and their detected versions on the authorized target, run:

```
nmap -sV TARGET_IP
```

9. Other Useful Commands

Purpose	Command
Basic scan	nmap TARGET_IP
Selected ports	nmap -p 22,80,443 TARGET_IP
Port range	nmap -p 1-1000 TARGET_IP
Service/version detection	nmap -sV TARGET_IP
OS detection*	nmap -O TARGET_IP
Save output	nmap TARGET_IP -oN scan.txt

* OS detection may require appropriate privileges and may not always identify the operating system accurately.

10. Understand the Scan Output

Field	Meaning
PORT	Network port number and protocol, such as 80/tcp.
STATE	Whether the port is open, closed, or filtered.
SERVICE	The service commonly associated with the port.

11. Procedure

1. Install Nmap on the system.
2. Open Command Prompt, PowerShell, or Terminal.
3. Identify the IP address of the authorized target.
4. Run **nmap TARGET_IP**.
5. Observe the discovered ports and services.
6. Run **nmap -sV TARGET_IP** if service/version detection is required.
7. Save the result using **-oN scan.txt**.
8. Document the scan findings.

12. Result

The Nmap scan was successfully performed on the authorized target, and the available port and service information was identified and documented.

13. Conclusion

Thus, a basic Nmap network scan was performed successfully and the discovered port and service information was analyzed and documented.

14. Quick Flow

Install Nmap → Find Target IP → Open Terminal → Run Nmap Scan → Analyze Ports/Services → Save Results → Document Findings

Submission note: Replace TARGET_IP with the IP address used in your authorized lab and attach the actual scan-output screenshot if required.